

Highlights

Geometric and Topological Data Analysis of Atomization Simulations for Model Development

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- A novel method to extract droplet geometries and flow fields from atomization simulations is implemented.
- The DIWECT is described and applied to atomization data.
- It is shown that complex geometric data from simulations can be vectorized while retaining key physical characteristics.

Geometric and Topological Data Analysis of Atomization Simulations for Model Development

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ABSTRACT

High-fidelity simulations of atomization present a promising approach to studying its complex physics. However, these simulations face significant challenges, often requiring hundreds of processors and generating tens to hundreds of terabytes of data, complicating their practical application. This work presents a method to extract localized geometries of droplets and their surrounding flow fields at the moment of breakup within a simulation, resulting in a comprehensive database of atomization statistics. This improves the usefulness of simulation results by storing key information on the atomization process. Parsing and comparing these geometries remains challenging, so we apply the directionally integrated weighted Euler characteristic transform (DIWECT) to produce vectorized representations of the complex data. These vectors are well-suited for advanced data processing, machine-learning applications, and the development of reduced-order models. We demonstrate the effectiveness of this approach through both shape matching and K-means clustering to confirm that the DIWECT captures meaningful physical information in an abstract, reduced-order format. This work represents a novel, efficient, and physically relevant method with which to extract, parse, and analyze large-scale atomization simulation data with an eye toward reduced-order model development.

1. Introduction

Atomization—the process where a coherent liquid body breaks into smaller droplets—is a wide-ranging phenomenon with applications in numerous industries and environmental applications. From fuel injection and fire suppression to agricultural spraying and air-sea interactions, a detailed understanding of the underlying physics is essential for optimizing engineering designs and improving predictive models of natural phenomena. Despite its widespread importance, many of the underlying physical mechanisms that drive atomization remain unresolved. There remains significant research in the field attempting to answer the fundamental questions: when and how do liquids breakup?

Experimental studies on atomization have long faced limitations due to the inherent complexity of capturing high-speed, multiphase flows. Simplified jet flows can be captured well using traditional high-speed imaging techniques Marmottant and Villermaux (2004). However, as the Weber and Reynolds numbers increase to values relevant for many practical applications, experimental analyses of the physics become intractable because of the formation of an opaque cloud of droplets that forms around the spray. Even with recent advances in imaging techniques like ballistic Linne, Paciaroni, Gord and Meyer (2005); Sedarsky, Berrocal and Linne (2010); Slangen, Lauret, Heymes, Aprin and Lecysyn (2016) and x-ray imaging Wang, Im, Fezzaa, Lee, Wang, Micheli and Laub (2006); Heindel (2018); Li, Zhao and Wu (2021), capturing temporally continuous, quantitative data remains problematic. As a result, experimental methods have struggled to provide sufficient insight into the small-scale instabilities and breakup mechanics that occur in atomizing systems.

In response to these challenges, high-fidelity simulations—commonly known as direct numerical simulations (DNS)—have emerged as a powerful alternative Gorokhovski and Herrmann (2008) capable of resolving the detailed physics of atomization. High-fidelity simulations continue to produce increasingly accurate representations of the complex multiphase process, providing researchers with a tool to predict quantitative, three-dimensional, and temporally granular descriptions of liquid breakup. However, these simulations are not without their own limitations. As

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computational power has increased, the size of datasets generated from atomization simulations has grown dramatically. High-resolution simulations, capable of producing hundreds of terabytes of data, present immense challenges in both storage and analysis. Extracting meaningful physics from such massive datasets remains an ongoing challenge, and current methodologies are ill-equipped to fully utilize this wealth of information.

To address these limitations, the Atomization Simulation Statistics Extraction Tool (ASSET) Christensen and Owkes (2023); Rubel and Owkes (2019) was developed to extract detailed event-based data from large-scale simulations of atomizing liquids. Unlike traditional post-processing methods that are constrained by the sheer volume of data, ASSET works coincident with a simulation to identify and record breakup and coalescence events as they occur. This approach allows for the creation of comprehensive databases that contain statistics, geometries of liquid structures, and the corresponding flow fields at the moment of each breakup event.

Furthermore, ASSET tracks the evolution of each liquid structure, allowing for the reconstruction of breakup *pathways*. These are the sequential processes by which liquid structures fragment into progressively smaller droplets. Breakup pathways reveal how individual droplets emerge from larger structures, shedding light on the mechanisms that govern spray formation. By capturing these data directly from simulations, researchers gain an unprecedented level of access to temporally continuous, localized information essential for studying the full physics of atomization.

High-fidelity simulations are incredibly costly, often requiring hundreds of thousands of CPU hours on high-performance computers. Thus, predictive atomization models play a vital role, especially for practical applications where fully resolving the physics is not feasible. Reduced-order models such as Reitz-Diwakar (RD) Reitz and Diwakar (1987), Taylor Analogy Breakup (TAB) O'Rourke and Amsden (1987), and Pilch-Erdman (PE) Pilch and Erdman (1987) models offer relatively fast and inexpensive ways to predict atomization behavior in specific systems. However, these models are heavily based on simplified assumptions and empirical data from idealized conditions, such as single droplets in controlled environments. As a result, they often fail to accurately represent the complex, multiscale behavior of atomization in real-world, turbulent systems. There is an increasing need for models that are informed by statistically relevant physics-based data extracted directly from simulations, enabling more accurate predictions of atomization dynamics across a broader range of conditions.

To meet this need, an essential step forward involves not only extracting more comprehensive data from simulations but also transforming these data into a format suitable for modern analysis techniques. Concretely, it is necessary to efficiently query and parse databases full of data from physical simulations. The present work proposes using a geometric and topological data analysis tool known as the directionally integrated weighted Euler characteristic transform (DIWECT) to convert complex geometric and flow field data from atomization simulations into vectorized representations. This approach structures the data in a format well suited for machine learning and other data-driven techniques, enabling the development of reduced-order models that agree with the physics of atomization predicted by high-fidelity simulations. The resulting structured database enables efficient querying and analysis, supporting the development of reduced-order models that retain the underlying physics of atomization rather than relying on significant assumptions.

This paper focuses on demonstrating the process of generating these databases and applying DIWECT to enable future data-driven predictive models. Furthermore, the DIWECT provides a powerful foundation for novel analyses, enabling the identification of previously unrecognized patterns within these large-scale, complex systems. In doing so, we aim to bridge the gap between the vast amounts of data produced by high-fidelity atomization simulations and the limited models currently used, advancing toward a more accurate and physically informed understanding of atomization.

1.1. Background - Topological Data Analysis

Modern experimental and numerical methods in all range of scientific fields are producing increasingly complex and large-scale data, requiring efficient methods to analyze and visualize them. Topological data analysis (TDA) methods have emerged as a family of tools for summarizing and analyzing data in a mathematically robust and rigorous manner. For a comprehensive background on the various methodologies see *Topological Methods in Data Analysis and Visualization VI* Hotz, Masood, Sadlo and Tierny (2021). In the context of fluid mechanics, where massive datasets are a common challenge, TDA offers the ability to capture key geometric components of complex fields.

There exist a few studies incorporating TDA into fluids Gameiro, Mischaikow and Wanner (2005); Krishan, Kurtuldu, Schatz, Gameiro, Mischaikow and Madruga (2007); Levanger (2017); Afanador, Zaleski, Tryggvason and Lu (2021); Nauleau, Bridel-Bertomeu, Vivodtzev, Beaugendre and Tierny (2021); Núñez, González and Ramos (2022). Scalar fields in fluid mechanics, such as pressure, temperature, and image pixel intensity can be analyzed using the

persistent homology TDA method. These analyses have been used to identify large-scale symmetries, transition regions, and patterns in the flow Kramár, Levanger, Tithof, Suri, Xu, Paul, Schatz and Mischaikow (2016); Roy, Haque, Mitra, Tarafdar and Dutta (2023). Similarly, studies involving vector fields, such as velocity, have used topological descriptors to classify coherent flow features and reveal interactions within turbulent flows Hasslberger, Ketterl, Klein and Chakraborty (2019a); Hasslberger, Marten and Klein (2019b). While these applications illustrate the versatility of TDA in fluid mechanics, they are focused on scalar field data and large-scale flow dynamics rather than capturing the small-scale, underlying physics of liquid breakup studied in this work.

In contrast, the geometric analysis of three-dimensional liquid structures in multiphase flows through TDA remains a relatively underexplored area. Chen et al. Chen, Lu, Zaleski and Tryggvason (2022) introduced a skeletonization technique to capture the fundamental topology of periodic liquid jets during atomization. This method allowed for the identification of branching points in the jet structure and offered new insights into the mechanisms driving liquid breakup. [Very recent work has been continued in this field describing breakup morphology using topological skeletons](#) Garcia and Ling (2026). Beyond fluid mechanics, TDA has been successfully employed in fields such as medical imaging He, Hong, Chen and Liang (2001); Pizer, Gerig, Joshi and Aylward (2003); Crawford, Monod, Chen, Mukherjee and Rabadán (2020), 3D object recognition Hilaga, Shinagawa, Kohmura and Kunii (2001), and structural biology Turner, Mukherjee and Boyer (2014) to quantify, match, and analyze complex shapes. These examples demonstrate the versatility of geometric TDA in capturing intricate structural information and motivate its application to fluid systems.

Building on these developments, this work focuses on analyzing three-dimensional geometries associated with liquid structures during atomization. Concretely, we use a topological tool referred to as the DIWECT for this purpose. Unlike previous methods that emphasize whole flow fields, the present work applies TDA to isolated liquid breakup events, providing concise, representative descriptions of complex geometries.

While other TDA techniques could be applied similarly to three-dimensional geometries, the DIWECT offers clear advantages in both interpretability and computational cost. Persistent homology, for example, provides similar topological summaries. However, computing persistent homology generally scales cubically with data size compared to linearly for the DIWECT. Additionally the results of persistent homology are not naturally vectorized, which is a primary motivation of this work. Similarly, the Mapper algorithm depends on multiple user-defined parameters, such as filter choice, overlap, and clustering method, all of which strongly influence the resulting topology. Mapper is also less easily parallelized and therefore scaled. The DIWECT requires only the number of sample directions and heights and it is fully parallelizable. These characteristics make DIWECT particularly well suited for the current problems that rely on scalability, consistency, and efficiency with large data.

1.2. Background - Atomization Modeling

One of the primary objectives of this work is to provide physical data that can inform reduced-order atomization models. Achieving this requires an understanding of the information needed and the limitations of current modeling approaches. This section focuses on the role of Lagrangian droplet tracking models, as the data extracted from this study is particularly suited to their improvement.

The Lagrangian-Eulerian (LE) technique is a method used to model atomization with reduced computational cost. It represents the liquid phase as discrete, Lagrangian “blobs” or droplets, while the gas phase is treated as an Eulerian domain. Prominent implementations, such as the KIVA series Amsden, Ramshaw, O’rourke and Dukowicz (1985); Amsden, O’Rourke and Butler (1989); Amsden (1993, 1997); Torres and Trujillo (2006), have been widely adopted in industry U.S. Department of Energy (2006). Following injection, the breakup of blobs is typically modeled using idealized instability mechanisms, such as Kelvin-Helmholtz and Rayleigh-Taylor instabilities Beale and Reitz (1999); Reitz and Diwakar (1987); Patterson and Reitz (1998). Further breakup is often predicted using models like the Taylor Analogy Breakup (TAB) O’Rourke and Amsden (1987) or Pilch-Erdman (PE) Pilch and Erdman (1987). These methods offer computational efficiency but rely heavily on simplifying assumptions, such as spherical droplet shapes and empirical correlations from isolated experiments.

These simplifications fail to capture the multiscale and turbulent nature of real-world atomization processes. Experiments and DNS have revealed that atomization involves complex deformation and breakup mechanisms, including bag and sheet formation, ligament destabilization, and droplet fragmentation Marmottant and Villermaux (2004); Jarrahbashi and Sirignano (2014); Zandian, Sirignano and Hussain (2018). These mechanisms lead to highly variable, complex droplet and flow field geometries that are far beyond the capabilities of traditional blob injection models.

Recently, there has been considerable development on the hybrid Volume-of-Fluid Lagrangian-Eulerian (VOFLE) method (Tomar, Fuster, Zaleski and Popinet (2010); Herrmann (2010); Zuzio, Estivalèzes and DiPierro (2018); Evrard, Denner and van Wachem (2019); Kuo and Trujillo (2022)). These models combine the Volume-of-Fluid (VOF) technique to resolve near field jet deformation and large-scale liquid structures before transitioning to an Lagrangian description for smaller droplets (Note: Level-set (LS) is also used for interface capturing in some cases (Zuzio et al. (2018))). Varying local criteria are used to determine when a droplet meets the requirements for Lagrangian treatment, with most opting for a size threshold for the drop diameter, and others additionally considering the sphericity of the droplet. Only two papers have considered further breakup to the author's best knowledge. These are Yu et al. (2017), who consider the RT instability for breakup and Kuo & Trujillo (2022), who use the maximum entropy formalism (MEF) model. VOFLE balances computational efficiency and accuracy with an improved resolution of the early stages of jet formation and breakup.

Despite these advancements, VOFLE models face challenges associated with empirical assumptions and computational trade-offs. For example, most methods require that droplets be spherical and/or below a size threshold before transitioning to a Lagrangian representation. This increases the necessary size and cost of the Eulerian domain. Additionally, secondary atomization is largely neglected, and when it is considered, secondary atomization models are significantly limited.

This work lays the foundation for a predictive VOFLE methodology, informed by high-resolution simulations. By extracting explicit breakup cascades, droplets can be transformed to Lagrangian structures at larger and less spherical sizes and their subsequent breakup can be more accurately predicted. The envisioned approach would identify liquid structures as they detach, compare them to a database of high-fidelity DNS results, and use stored breakup pathways to inform how a given structure will fragment. While this study does not implement the full VOFLE model, it establishes techniques for extracting droplet geometries, surrounding flow fields, and breakup genealogies, which are necessary steps toward improving atomization modeling.

2. Methodology Overview

This work seeks to establish a novel method to bridge the gap between high-resolution simulations and reduced-order modeling of atomization processes. The ultimate goal is to use the data and processing techniques outlined in this paper to inform reduced-order atomization models. However, the development of a model is beyond the scope of the present work. The novelty of this work lies in (i) advancing the ASSET framework to robustly extract and organize breakup geometries and their surrounding flow fields, and (ii) introducing the DIWECT and, more broadly, geometric and topological data analysis of 3D shapes in multiphase flow. The current project is structured into these two key objectives:

1. **Enhance ASSET to extract atomization geometries:** The ASSET methodology is extended and optimized to extract the gas–liquid interface of liquid structures, along with their surrounding flow fields, from high-fidelity simulations during breakup and coalescence events. These structures represent a significant advancement to the original methodology described by Christensen and Owkes (2023).
2. **Vectorize the resultant databases using the DIWECT:** The geometric and flow field data extracted using ASSET are transformed into vectorized representations with the DIWECT. This process prepares the data for machine learning applications, allowing for statistical analysis and efficient shape-matching. By capturing key geometric and topological features, the DIWECT helps identify patterns and relationships within the data.

This work provides an approach to analyze large atomization simulation databases, yielding information necessary to inform a physics-based reduced-order model. Explicitly, it is important to analyze the DIWECT as a method for vectorizing data in a *useful* way—where useful refers to the ability of these representations to serve as effective inputs to machine learning algorithms. An ideal vectorization must preserve the key geometric and topological characteristics of complex flow structures while reducing them to a comparable format suitable for statistical analysis and training. This ensures that meaningful similarities and differences between structures are captured in a way that algorithms can learn from and generalize across. This quality will provide both novel, in-depth, data-driven analyses into atomization as well as the potential to use these insights for predictive modeling.

3. Methods

3.1. Computational Platform

The advancements outlined below are built into the NGA computational platform (Desjardins, Blanquart, Balarac and Pitsch, 2008a; Desjardins, Moureau and Pitsch, 2008b; Owkes and Desjardins, 2014, 2017). This platform is then used to perform the atomization simulation outlined below. NGA solves the two-phase, incompressible Navier-Stokes equations for mass and momentum conservation, expressed as

$$\frac{\partial \rho_\phi}{\partial t} + \nabla \cdot (\rho_\phi \mathbf{u}_\phi) = 0 \quad (1)$$

and

$$\frac{\partial \rho_\phi \mathbf{u}_\phi}{\partial t} + \nabla \cdot (\rho_\phi \mathbf{u}_\phi \otimes \mathbf{u}_\phi) = -\nabla p_\phi + \nabla \cdot (\mu_\phi [\nabla \mathbf{u}_\phi + \nabla \mathbf{u}_\phi^T]) + \rho_\phi \mathbf{g}, \quad (2)$$

where ρ_ϕ is the density, $\mathbf{u}_\phi$ the velocity vector, p_ϕ the pressure, μ_ϕ the dynamic viscosity, t the time, $\mathbf{g}$ the gravitational acceleration vector, and ϕ the phase indicator (g for gas and l for liquid). These equations are discretized on a staggered Cartesian grid, pressure stored at cell centers and velocity components at cell faces.

Numerical methods include:

- Time discretization: An iterative Crank-Nicolson scheme, with a semi-implicit correction Choi and Moin (1994)
- Gas-liquid interface tracking: A geometric volume-of-fluid (VOF) method with unsplit semi-Lagrangian fluxes Owkes and Desjardins (2014). The interface is reconstructed using piecewise linear interface calculation (PLIC) Youngs (1982) with efficient least-squares VoF interface reconstruction algorithm (ELVIRA) for normal vector calculations Pilliod and Puckett (2004) on subcells to ensure mass and momentum conservation Owkes and Desjardins (2017).
- Advection: Away from the interface, high-order finite differences for mass, momentum, and scalar transport Desjardins et al. (2008a) are used. Near the interface the semi-Lagrangian VOF method is used Owkes and Desjardins (2014).
- Pressure Solver: The Poisson equation is solved with the ghost fluid method to incorporate the surface tension force Fedkiw, Aslam, Merriman and Osher (1999) and a black-box solver Dendy (1982).
- Surface Tension: Curvature is computed using the adjustable curvature evaluation scale (ACES) method Owkes, Cauble, Senecal and Currie (2018).

NGA is fully parallelized via MPI, enabling efficient scaling to tens of thousands of cores Desjardins and Pitsch (2010). Its computational efficiency and robust numerical methods have made it an effective tool for studying atomization and related multiphase flow phenomena (Desjardins, McCaslin, Owkes and Brady, 2013; Krolick and Owkes, 2018; Owkes and Desjardins, 2014; Sheehy and Owkes, 2017).

3.2. ASSET Methodology

The first goal in this project is to extract liquid structure and flow field geometries from high-fidelity simulations. To meet this goal, it is necessary to identify key moments and extract only those structures that provide model-relevant data. The breakup process is important because it describes the formation of droplets. However, breakup represents only part of the dynamics in large-scale atomization. Coalescence is also common in dense spray clouds. Thus, tracking both breakup and coalescence is necessary to fully describe the system.

In addition, liquid structures often deform significantly between breakup or coalescence events. These deformations provide information on the mechanisms leading to breakup, such as an elongating ligament undergoing Plateau-Rayleigh instability or a droplet flattening before undergoing bag-type breakup.

The Atomization Simulation Statistics Extraction Tool (ASSET) is designed to capture this information by identifying and isolating breakup, coalescence, and intermediate (update) events. ASSET is an event-triggered, parallelized algorithm that operates concurrently with simulations. It runs each timestep and efficiently identifies breakup and coalescence events, gathering physical data associated with each occurrence. A comprehensive description

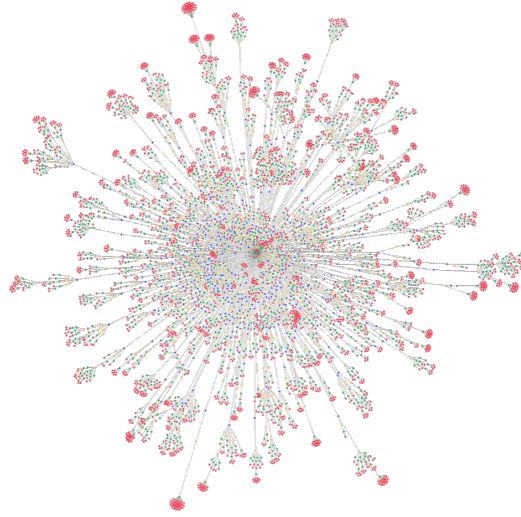


Figure 1: Neo4j graph database representing data extracted from a numerical simulation of a diesel-type jet from Christensen and Owkes (2023). Dots represent droplets with lines connecting related droplets (i.e parent and child droplets). Colors represent the number of breakups taken to get to a given droplet.

of the algorithm is given in the original works Rubel and Owkes (2019); Christensen and Owkes (2023), however, a concise summary is included here. As we detail below, the key addition in this work is the ability to extract topologically sound geometries and to incorporate update events for more comprehensive tracking of liquid structure evolution.

The workflow for data extraction involves three key steps: event identification, geometry extraction, and velocity calculation on the extracted meshes. 1) Event identification works by monitoring changes in identification numbers, providing liquid structure history and breakup pathways. 2) Once an event is detected, the geometries of the affected liquid structures are extracted by isolating the computational cells involved in the event. To generate a watertight surface representation, a marching cubes algorithm is applied, ensuring topological accuracy and avoiding artifacts that could interfere with subsequent analyses. 3) Finally, rather than storing the full velocity field, only the velocity component normal to the liquid surface is recorded, as this component is most relevant to droplet deformation. Details on each step are provided below.

3.2.1. Event Identification

Event identification in ASSET (described in detail by Christensen and Owkes Christensen and Owkes (2023)) begins by tagging liquid structures with two unique integers: S and $\mathcal{L}$. S is assigned at every timestep using a band-growth algorithm Herrmann (2010), distinguishing independent liquid structures within the domain. $\mathcal{L}$ is persistent across timesteps and is transported with the liquid, only changing when breakup or coalescence occurs. $\mathcal{L}$ provides a history of the liquid, allowing us to track the breakup pathway of any given liquid structure.

The event identification algorithm distinguishes the three types of events as follows:

- **Breakup Events:** Occur when a single liquid structure (same $\mathcal{L}$) splits into multiple, structures (different S) in non-adjacent grid cells.
- **Coalescence Events:** Identified when two or more structures with distinct $\mathcal{L}$ values flux into a single grid cell.
- **Update Events:** Triggered periodically for each structure based on a user-defined time interval relative to the last event the structure was involved in.

The original method described the extraction of statistics when an event was identified. This utility is still applied. These data are sampled both immediately preceding and following an event, providing a record of the conditions leading to and resulting from each event. Data are stored in a Neo4j graph database (see Figure 1), enabling relationships between parent and child droplets to be established and liquid structures to be tracked back to the initial coherent liquid at the start of the simulation.

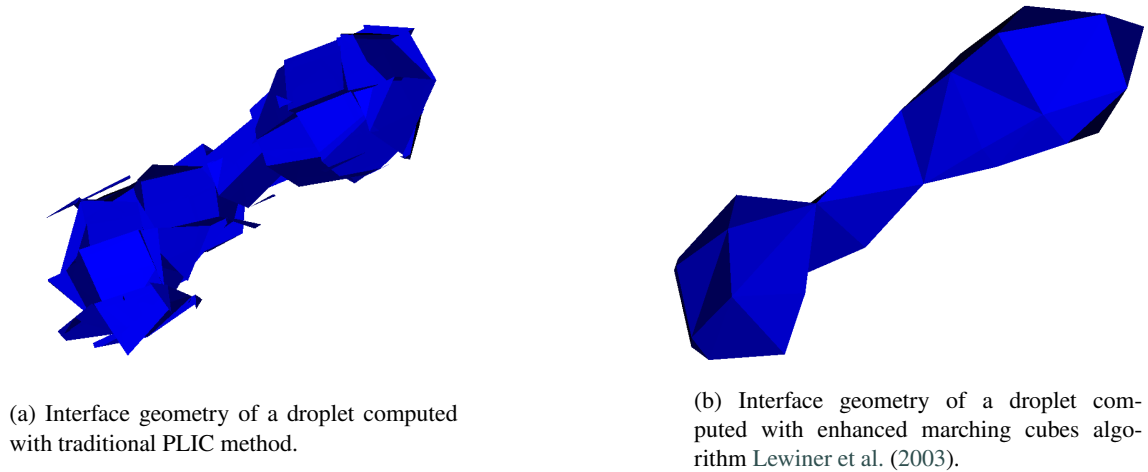


Figure 2: Comparison of the volume conserving PLIC interface and the topologically sound, watertight marching cubes interface.

3.2.2. Geometry Extraction

When an event is identified, the region around the liquid structure(s) in the event is isolated. In fact, this is already done during the assignment of S through a band-growth algorithm Herrmann (2010), providing access to the computational cells of interest. With the cells isolated, the liquid-gas interface geometry is captured. Ensuring no erroneous topological features are captured in this step is necessary because the DIWECT methodology is highly sensitive to them. A naive implementation would be to output the PLIC representation of the interface, which is already calculated. However, as seen in Figure 2a, the edges and vertices of the PLIC interface are not aligned to produce a watertight surface, introducing non-physical holes. To ensure topological soundness, we implemented a formulation of the commonly used marching cubes algorithm Lorensen and Cline (1987) that is topologically robust Lewiner, Lopes, Vieira and Tavares (2003). This method generates a set of triangles representing the interface. The vertices and connectivity information of these triangles are stored as an unstructured mesh in the SILO format Childs, Brugger, Whitlock, Meredith, Ahern, Pugmire, Biagas, Miller, Harrison, Weber, Krishnan, Fogal, Sanderson, Garth, Bethel, Camp, Rübel, Durant, Favre and Navrátil (2012). In this format, each triangle is represented as a zone, and scalar values can be associated with these zones. The marching cubes algorithm provides a topologically sound representation of the droplet, however, it does not ensure a fully accurate volume, whereas the PLIC interface is volume conserving, which is necessary for the geometric volume-of-fluid numerical methods. Producing a volume-conserving, water-tight representation and preventing the calculation of two separate interface reconstructions is an area of future work.

3.2.3. Velocity Calculation

Saving the full velocity field around each droplet would present a significant storage requirement. To reduce the total data being saved, the flow field is simplified to include only the velocity component normal to the interface. This normal component, represented mathematically as $\mathbf{u} \cdot \mathbf{n}$, where $\mathbf{u}$ is the velocity vector and $\mathbf{n}$ is the unit normal vector of the interface plane. *Storing the interface normal velocity retains the flow information most relevant to interface deformation, while limiting the significant data load associated with describing the full local flow field. This trade-off accomplishes the goal of a compact characterization of flow field with the highest retention of valuable information.*

Since $\mathbf{u} \cdot \mathbf{n}$ is a scalar quantity, it can be efficiently stored in the SILO file for each triangle zone. With the orientation of the interface plane and the velocity magnitude, the interaction between the velocity and the shape of the fluid is efficiently stored together. The simplified data can then be visualized and analyzed using tools such as VisIt Childs et al. (2012), enabling detailed studies of droplet dynamics.

3.3. The Directionally Integrated Weighted Euler Characteristic Transform (DIWECT)

With droplet geometries and their associated normal velocities now extracted and stored efficiently in a compact, structured format, the next objective is to develop a means of comparing these complex shapes and dynamics in a consistent, scalable manner. Specifically, we require a vectorized representation that captures the essential geometric

and flow-related features of each droplet, enabling fast and meaningful comparison across a large database of breakup events. The Directionally Integrated Weighted Euler Characteristic Transform (DIWECT) is introduced here as a method to provide exactly this vectorization, translating high-dimensional surface and velocity data into forms suitable to statistical analysis and machine learning.

This section details the methodology of the DIWECT. This method is an extension of the weighted Euler characteristic transform (WECT). For a more in-depth background on the WECT, see Jiang et al. Jiang, Kurtek and Needham (2020) and Cisewski-Kehe et al. Cisewski-Kehe, Fasy, Giriyan and Quist (2023). The following passages describe the DIWECT in a mathematically rigorous way to ensure reproducibility and precision. However, the method is intended for use among a range of scientists and engineers. Thus, intuitive explanations are also provided to convey the underlying physical meaning of each mathematical definition. At a high level, the DIWECT takes a complex droplet surface along with the velocity field and describes it with a vector. Each element of the vector describes how the droplet's shape and normal velocity vary when viewed from different directions and at different depths into the shape.

The foundation of this method are simplicial complexes. Explicitly, for a set of vertices $\mathcal{V} \subseteq \mathbb{R}^n$, a *simplex* is the convex hull of an affinely independent subset of $\mathcal{V}$. Given any simplex σ , we define its dimension $\dim \sigma$ as the number of vertices in σ minus one (e.g. a zero-simplex is a vertex, a one-simplex is an edge, a two-simplex is a triangle, etc.). A face τ of a simplex σ is any subset of its vertices ($\tau \leq \sigma$), while σ is a coface of τ ($\sigma \geq \tau$). A *geometric simplicial complex* is a finite collection of simplices K satisfying:

1. If $\sigma \in K$, then all faces $\tau \leq \sigma$ are also in K
2. If any two simplices $\sigma, \sigma' \in K$, have a nonempty intersection, then $\sigma \cap \sigma'$ is a face of both σ and σ'

In practical terms, each droplet surface is represented as a mesh of interconnected triangles (faces), edges, and vertices that fully describe the geometry. The simplicial complex formalism ensures that this mesh can be analyzed consistently, regardless of how finely it is discretized.

A *weighted simplicial complex* (K, ω) is a simplicial complex K together with a weight function $\omega : K \rightarrow \mathbb{R}$, that assigns a scalar to each simplex. The unstructured meshes extracted in §3.2.2 are simplicial complexes with the normal velocities calculated in §3.2.3 as weights on each triangle. Weights for the edges and vertices are calculated by averaging the weights of their respective co-faces.

The Euler characteristic of a simplicial complex is given as

$$\chi(K) = \sum_{\sigma \in K} (-1)^{\dim \sigma}. \quad (3)$$

which, for a mesh of vertices, edges, and faces, reduces to the familiar $\chi = V - E + F$ where V , E , and F are the numbers of vertices, edges, and faces, respectively. For example, the Euler characteristic of any Platonic polyhedra (e.g. tetrahedron, cube, octahedron, etc.) is $\chi = 2$. The Euler characteristic is a topological invariant, meaning it remains unchanged under continuous deformations such as bending or stretching, as long as there is no tearing or gluing. In other words, χ acts as a scalar measure of the object's topology. It takes into account the number of parts or holes that exist in a structure.

The Euler characteristic is easily extended to the *weighted Euler characteristic* (χ_ω) ,

$$\chi_\omega(K, \omega) = \sum_{\sigma \in K} \omega(\sigma) \cdot (-1)^{\dim \sigma}, \quad (4)$$

in which each simplex contributes its weight rather than ± 1 . Effectively, the weighting works the normal velocity the description of the shape by determining how much each feature contributes to the final value. **Note that the weight function ω must be scalar-valued, so the full velocity vector cannot serve as a weight. The normal component is selected as the scalar most directly associated with interface motion (see §3.2.3).**

The weighted Euler characteristic alone is a limited representation of any complex, because it reduces all information to a single number. **More information can be retained by slicing the droplet with a moving plane and recording how χ_ω changes as the plane sweeps through. Formally, this sweep is a *height filtration*: for a unit direction d , the height $h_d(\sigma)$ of a simplex σ (single vertex, edge, or face of the mesh) is that of its highest vertex, and the subcomplex at sweep position t collects every simplex lying entirely at or below the plane,**

$$K_{d,t} := \{\sigma \in K \mid h_d(\sigma) \leq t\}. \quad (5)$$

As t increases, $K_{d,t}$ grows from an empty complex to the full mesh. Evaluating the weighted Euler characteristic at each sweep position defines the *weighted Euler characteristic function* for that direction,

$$\text{WECE}_{(K,\omega),d}(t) := \chi_\omega(K_{d,t}, \omega), \quad (6)$$

which records how the droplet's topology and interface-normal velocity accumulate in direction d . See Figure 3 for a demonstration of this process along a single direction.

Repeating this sweep over many directions defines the *weighted Euler characteristic transform*,

$$\text{WECT}_{(K,\omega)}(d, t) := \text{WECE}_{(K,\omega),d}(t). \quad (7)$$

In practice, sampling the function at discrete height values produces a vector for each direction whose entries are the weighted Euler characteristic at those heights. Performing the same sweep across all directions produces a collection of these vectors.

Theoretically, the WECT fully encodes the shape of the simplicial complex K when computed over all directions and heights; with sufficiently dense sampling, it retains enough information to uniquely characterize a given complex Jiang et al. (2020); Cisewski-Kehe et al. (2023). In principle, this means the droplet geometry and its normal-velocity distribution could be reconstructed exactly from a sufficiently sampled WECT.

The WECT is sensitive to rotations and reflections of the simplicial complex. To remove this dependence, we introduce the *directionally integrated WECT*, or DIWECT which averages the WECT over the sampled directions. For a uniformly sampled set of directions $\mathcal{D}$, the DIWECT is estimated as

$$\text{DIWECT}_{(K,\omega)}(t) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \text{WECT}_{(K,\omega)}(d, t). \quad (8)$$

Averaging over the directions removes dependence on droplet orientation: Two droplets that have identical shapes and velocity fields but rotated differently will have the same DIWECT.

In practice, the WECT cannot be computed over all directions and heights, so we use a discretely sampled estimate and the theoretical guarantees are necessarily limited by the sampling. For this study, we selected 150 directions and 75 heights to balance two competing objectives: preserving fidelity of the droplet representation and minimizing storage requirements. As shown in Equation 8, the directional sampling is averaged over each height, yielding a 75-dimensional vector per droplet. This dimensionality was chosen to ensure that the resulting representation remains compact and do not increase the storage requirement for small droplets. For example, an octahedron, which is among the most simplified and lowest-resolution shapes extracted in our dataset, requires approximately 50 integers to represent its vertices, faces, and associated velocity components. Most droplets in the dataset are significantly more complex than an octahedron, so a 75-element vector does not substantially increase the storage burden for the majority of cases, while still providing a decently high-resolution description of shape variation across directions. Moreover, since the DIWECT of a single orientation is sufficient to represent any rotated or reflected transformation of a droplet, we avoid the need to store multiple transformed variants, effectively reducing the total storage cost by a factor of $\mathcal{D}$, the number of sampled directions. The dimensionality and storage requirements of the DIWECT scales linearly with the number of sampled directions and heights. A full analysis of the performance based on the number of elements will be the focus of future work.

Considering the DIWECT applied to simulation results, it is natural to consider the effect of grid resolution of the extracted geometries on the resulting DIWECT vectors. Because the DIWECT is a topological descriptor, it is only sensitive to features such as connected components or holes. As long as the grid resolution is sufficient to capture these features in their correct locations, the resulting DIWECT remains a reliable descriptor of the droplet. For example, a perfect sphere and any closed, convex polyhedron both have an Euler characteristic of 2, so refining the geometric resolution (e.g., from a coarse representation of a sphere, such as a single tetrahedron, to an exact sphere), the Euler characteristic remains the same. This is not to say that the DIWECT of a tetrahedron and a sphere would be the same, however, because the DIWECT depends on the geometric distribution of surface features. As long as those surface features are sufficiently well resolved, they are inherently captured in the DIWECT calculation through the Euler characteristic contributions that make up each vector element. It is well known that in the field of CFD in general, ensuring all interface features are captured accurately in a simulation remains an intractable problem for many complex flows. However, the methodologies presented here are intended to scale with improvements in simulation fidelity, providing a consistent and reliable method for analyzing this type of data.

Algorithm 1 Methodology Summary

```

1: Run Simulation and Extract Data with ASSET
2: while simulation is running do
3:   Assign initial  $S$  and  $\mathcal{L}$  identifiers in the domain
4:   for each timestep do
5:     Reassign  $S$  for each liquid structure
6:     Transport  $\mathcal{L}$  with the liquid
7:
8:     Event Identification:
9:     if Multiple  $S$  associated with one  $\mathcal{L}$  then
10:       Breakup event identified
11:     end if
12:     if Multiple  $\mathcal{L}$  in one cell then
13:       Coalescence event identified
14:     end if
15:     for each structure do
16:       if (Current time % update frequency = 0) then
17:         Update event identified
18:       end if
19:     end for
20:
21:     Geometry Extraction:
22:     for each identified event do
23:       Extract the gas–liquid interface via marching cubes
24:       for each interface cell do
25:         Velocity Calculation:
26:         Compute  $\mathbf{u} \cdot \mathbf{n}$ , on each triangle of the mesh
27:       end for
28:       Save as unstructured mesh with normal velocities
29:     end for
30:   end for
31: end while
32:
33: Post-Processing: Estimate the DIWECT
34: Aggregate all extracted data
35: for each unstructured mesh  $\chi$  do
36:   Estimate the DIWECT of  $\chi$ 
37: end for
38: Output: A vectorized representation of liquid structure geometries from atomization

```

4. Results

The results presented here are derived from $\approx 120,000$ droplets extracted from a high-fidelity simulation run. These droplets represent all extracted instances as described in §3.2.1, i.e. they are the result of breakup, coalescence, and update events. The entire simulation database, made up of a full rendering of each timestep requires $\approx 10\text{TB}$ of storage space. The total storage requirement of the unstructured SILO meshes was 133GB. The full WECT was calculated with 150 directions and 75 heights and only required 1.5 GB of storage. The averaging step was conducted over each direction and the DIWECT representation of the same dataset only required 10 MB—a reduction of over 13,000-fold from the SILO meshes and 1,000,000-fold from the original simulation dataset.

The following subsections provide proof of concept of the DIWECT as a method to uniquely and efficiently characterize liquid shapes and surrounding flow fields. First, we match droplets to their most similar counterparts

Table 1

Non-dimensional parameters for the simulation. D is initial droplet diameter, dx is grid spacing, ρ_g is gas density, U_g is the bulk gas velocity, μ_l is liquid viscosity, σ is the surface tension coefficient

Parameters	Description	Value
Cells across diameter	D/dx	150
Bulk Reynolds	$\rho_g U_g D / \mu_l$	4.74×10^4
Bulk Weber	$\rho_g U_g^2 D / \sigma$	825

using only their DIWECT values. Then, we show that the droplets can be clustered into groups which display physical relevance based on their DIWECTs.

4.1. Simulation Setup

To test the methodology, we ran a large-scale numerical simulation of an atomizing water droplet (Fig. 4a). The methods outlined in this work are designed to be scalable and capable of processing large numbers of droplets, as would be expected in most practical atomization applications. The setup in this work consists of a high-speed gas inflow at the x^- wall, which is added to an initialized homogeneous isotropic turbulence (HIT) domain and inflow, as shown in Figure 4b. This simulation ran for one month on 128 processors on the Tempest HPC at Montana State University. To reduce the size of the resultant dataset, only droplets from the second half of the simulation were analyzed. [Droplets are tracked by ASSET until they leave the domain through any wall and every breakup, coalescence, and update event is stored.](#)

This test case represents a common atomization scenario with broad applications, including aerial firefighting, agricultural spraying, industrial spray cooling, and environmental aerosols. The key non-dimensional parameters for this simulation are listed in Table 1. [The resolution of 150 cells across the initial diameter is comparable to recent high-fidelity atomization studies \(Meng and Colonius, 2018; Dorschner, Biasiori-Poulanges, Schmidmayer, El-Rabii and Colonius, 2020\).](#) Nevertheless, [at the present Weber number, breakup occurs at a wide range of scales producing secondary droplets that likely rely on unresolved sub-grid breakup.](#) As with any DNS approach, this simulation is presented as one numerical realization of the atomization process rather than a complete representation of all range of breakup mechanisms. We emphasize that this does not affect the validity of the methods presented in this work. Shape-matching and clustering analyses are comparisons within the single presented dataset. The presented methodology operates identically at any resolution. Unresolved small-scale breakup influences the population statistics of the resulting database, but not the validity of the extraction or vectorization.

4.2. Droplet Matching

A key objective of this work is to establish a basis for efficiently comparing droplets and their local flow fields across large simulation databases. In a future scenario, a user might extract a particular droplet of interest from an ongoing simulation and automatically search a previously computed database for the most similar droplet. Then, predictions can be made from the previously simulated droplet to inform how the droplet of interest might behave.

For demonstration, we selected a set of four droplets from a separate simulation run. These droplets were pulled from another water drop atomization case, but with a larger initial diameter. Four common shapes from the database were chosen. Each droplet's DIWECT signature was then compared against the database developed from the simulation in §4.1 to find the most similar entries according to Euclidean distance in DIWECT-space. These results can be seen in Figures 5-8, which show the reference droplet on the left, and the closest matched droplets on the right. AR is the aspect ratio, defined as the shortest principal axis divided by the longest, and V , which is the volume of the droplet.

The matching results show very nice qualitative agreement for the shapes. There exists some variation between the shape parameters, but it is important to note that there are many interacting factors that lead to the final DIWECT calculation, notably, the DIWECT captures information about the normal velocity components. Thus, comparing a simple shape parameter and the size of droplets is a limited assessment of the DIWECT's capability of to compare similar structures.

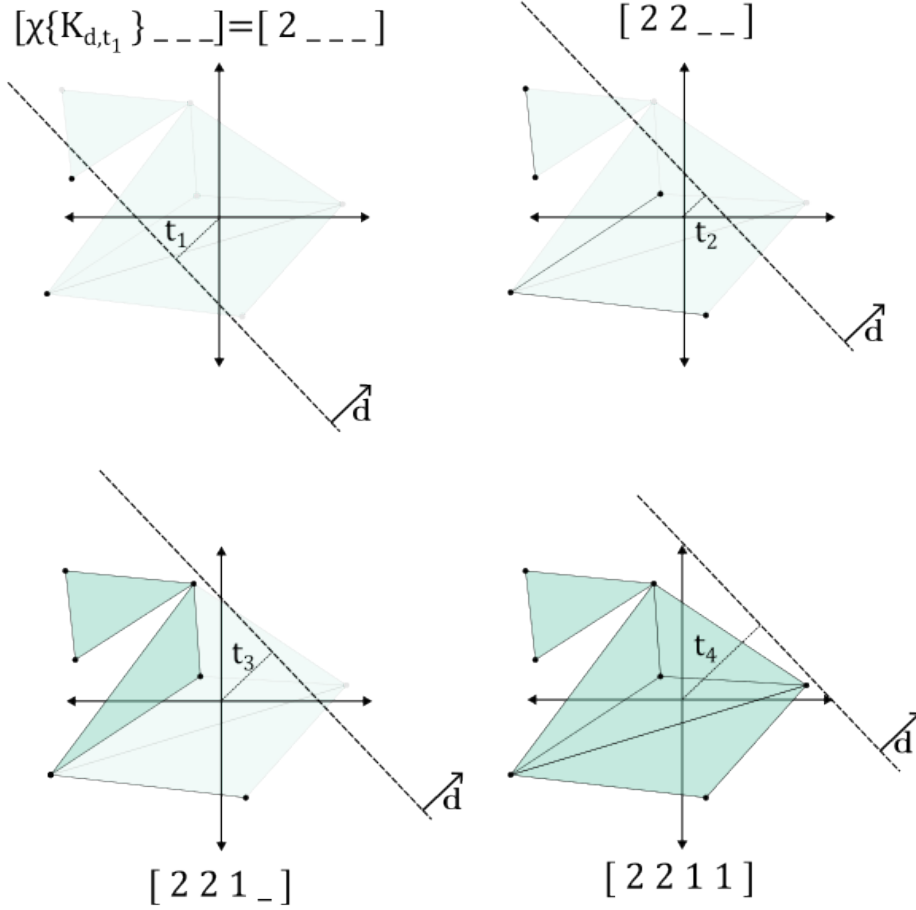
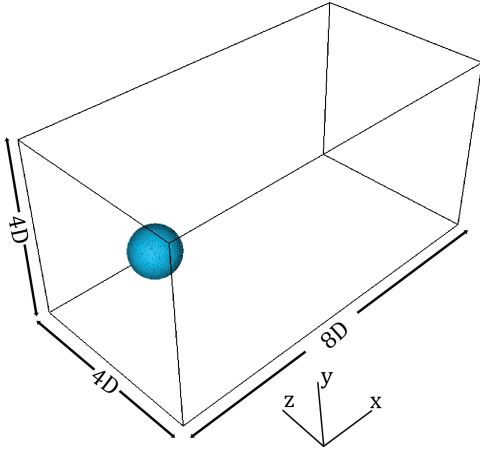
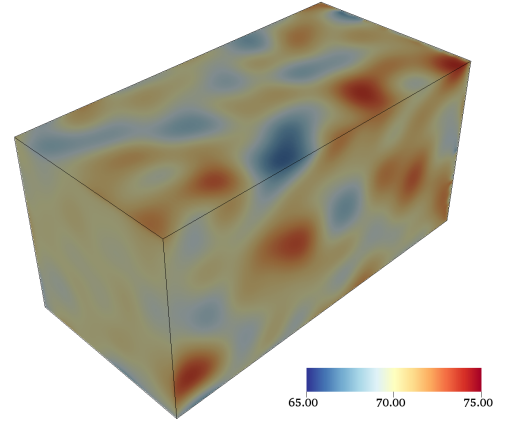


Figure 3: Euler characteristic χ computed over a height filtration of a two-dimensional example complex, in a single direction d at four heights $t_1, \dots, t_4$. Simplices become bold once their highest vertex lies below the sweep plane, at which point they enter the subcomplex $K_{d,t}$. The subcomplexes and their Euler characteristics, where V , E , and F denote the numbers of vertices, edges, and faces, are $K_{d,t_1} = (2V, 0E, 0F)$, $\chi = 2$; $K_{d,t_2} = (5V, 3E, 0F)$, $\chi = 2$; $K_{d,t_3} = (6V, 7E, 2F)$, $\chi = 1$; $K_{d,t_4} = (7V, 11E, 5F)$, $\chi = 1$. This sequence of χ values is the Euler characteristic function for direction d ; computing it over all sampled directions yields the Euler characteristic transform, and assigning weights to each simplex extends it to the weighted Euler characteristic transform (WECT) of Equation 7.

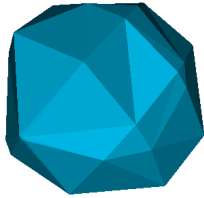


(a) Simulation domain with initial droplet in place (D is droplet diameter).



(b) HIT-derived turbulence initialized in the domain.

Figure 4: Simulation domain setup.



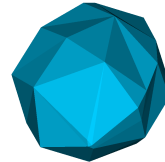
(a) Droplet 1: $AR = 0.868$, $V = 3.3 \times 10^{-12}$



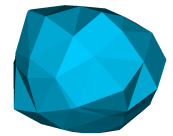
(b) $AR = 0.703$, $V = 5.22 \times 10^{-12}$



(c) $AR = 0.782$, $V = 2.99 \times 10^{-12}$



(d) $AR = 0.755$, $V = 5.74 \times 10^{-12}$



(e) $AR = 0.780$, $V = 1.44 \times 10^{-11}$

Figure 5: Droplet 1 and the four closest matches based on DIWECT

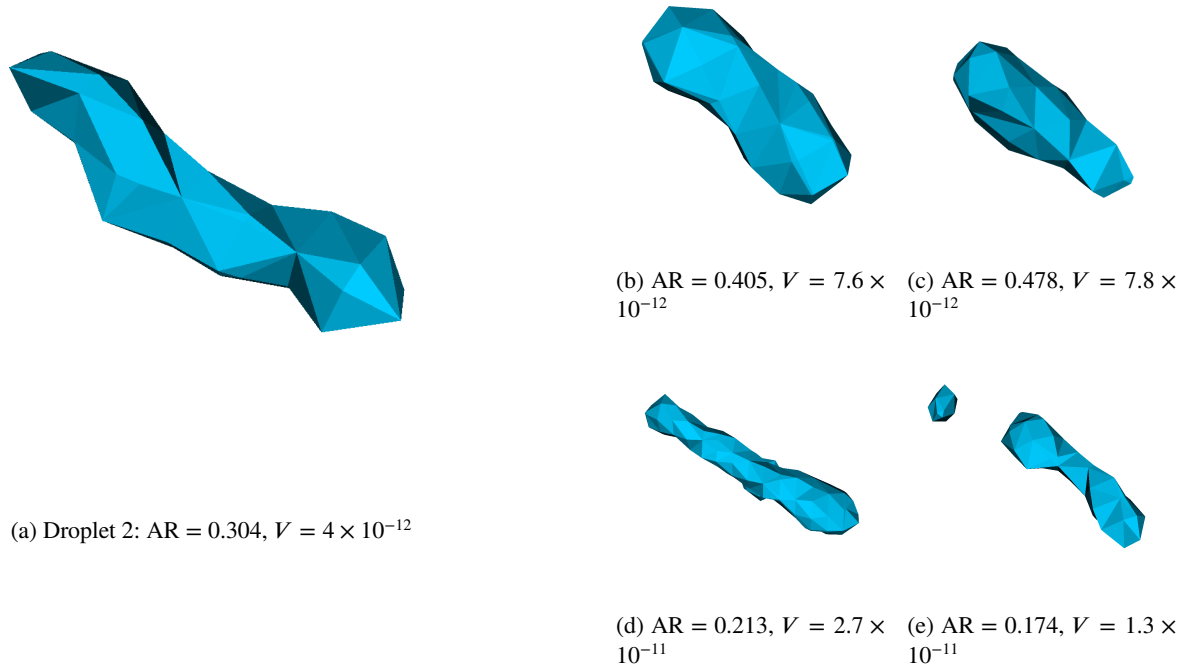


Figure 6: Droplet 2 and the four closest matches based on DIWECT

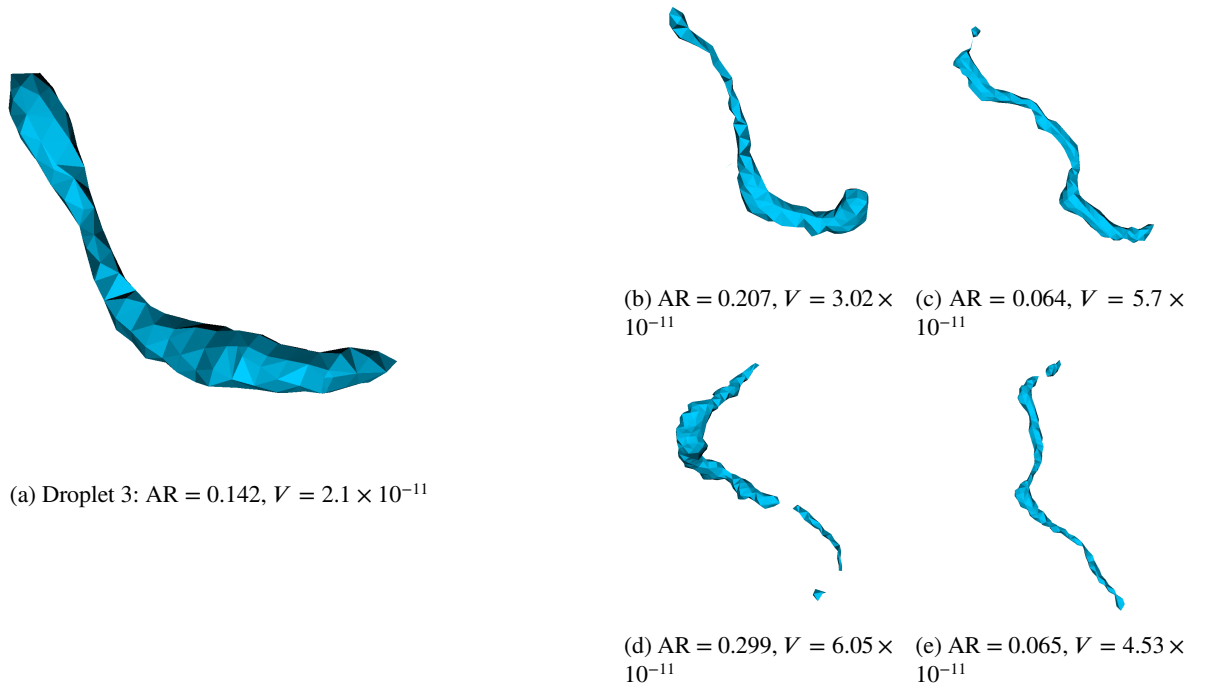


Figure 7: Droplet 3 and the four closest matches based on DIWECT

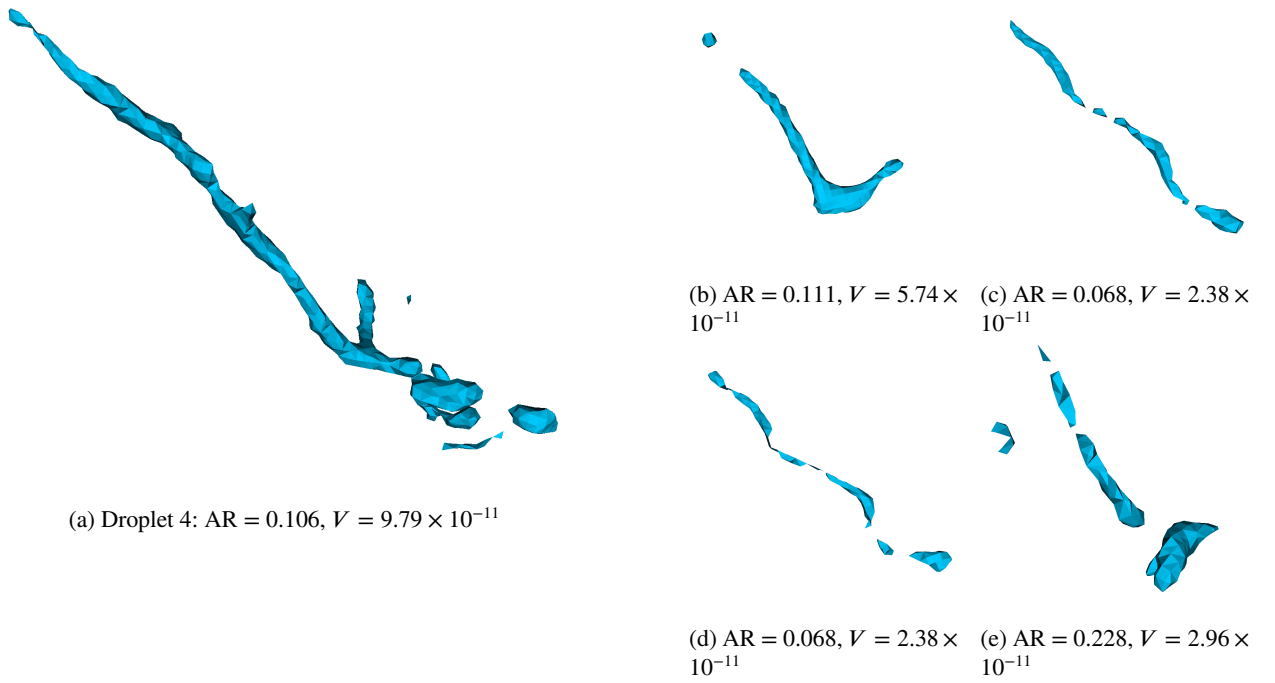


Figure 8: Droplet 4 and the four closest matches based on DIWECT

4.3. Clustering Results

Clustering serves as a practical demonstration of how the extracted geometries and their DIWECT representations can be used to quantify the wide range of droplets that are created in a spray. Due to the massive quantity of droplets created in most practical sprays, this type of data benefits from clustering the data into representative groups. Grouping droplet morphologies from high-fidelity simulations has not been previously demonstrated to our knowledge, but it is now possible through the extracted geometries and the DIWECT.

To demonstrate this capability, we apply clustering to the resulting DIWECT vectors using Euclidean distance as a measure of similarity. By comparing the resulting clusters with known physical parameters, we assess how effectively the DIWECT preserves relevant physical information. Given the large size of the dataset ($\approx 120,000$ samples), computational efficiency is a key consideration. To ensure efficiency of our analysis at scale, we consider methods better than $O(n^2)$ where n is the number of samples. Another consideration is the dimensionality of the data. The discretization produces a database of 75-dimensional vectors, so the dimensionality must be considered to avoid sparse sampling in high-dimensional space, otherwise known as the curse of dimensionality. Projecting the data into a lower dimension improves the capability of non-parametric methods and enhances computational efficiency. Principal Component Analysis (PCA) is therefore applied to the dataset before clustering. We found that 20 components was sufficient to explain the variability of our data (See Figure 9). To determine the appropriate number of PCA components, we used the elbow method, which identifies the point beyond which additional components contribute diminishing returns in explained variance. Following the application of PCA, we cluster the data using the K-means algorithm Lloyd (1982). We choose K-means because of its scalability, with a complexity of $O(nki)$, where n is the number of droplets, k is the number of clusters, and i is the dimensionality after PCA. This methodology provides both computational efficiency and a systematic way to assess, in a quantitative manner, whether the DIWECT representation retains physical information from atomization systems.

We begin with relatively few clusters ($K = 4$) to illustrate how the droplets naturally separate into groups with similar morphologies. Figure 10 provides a random sampling of droplets from each cluster. Some droplets appear similar because they are either observed at consecutive update events (when only minor shape changes occur) or they represent large structures that experience a small breakup event (e.g. droplet shedding off of a ligament), which is not visible at the scale of the image.

From inspection of Figure 10, cluster 0 appears to contain small, elongate droplets, whereas cluster 1 has small but more spherical droplets. Clusters 2 and 3 generally contain larger, more complex shapes. A selection of statistics describing these clusters is shown in Figure 11, with mean values and standard deviations indicated by error bars. The quantities plotted—volume, aspect ratio, slip velocities, and cluster population—capture important physical characteristics of the breakup process. Aspect ratio is defined as the ratio of the smallest principal axis length to the largest principal axis length (L_{min}/L_{max}); lower values indicate more elongate shapes, while higher values indicate more spherical droplets. Slip velocities refer to the difference between the droplet velocity and the surrounding gas velocity in the x-, y-, and z-directions, which influences the aerodynamic forces acting on the droplet. [These components are defined in the simulation reference frame with x aligned with the bulk gas inflow, such that \$U_{slip}\$ represents the streamwise slip.](#) The slip velocity is a slightly different (though related) quantity than the normal components of the velocity used as weights in the DIWECT. The slip velocity is calculated as a weighted sum of the gas velocity upstream of the droplet, which should represent the portion of the flow that impinges on the droplet.

Figure 11 shows that clusters 0 and 1 are composed of large numbers of small droplets; cluster 0 tends to have more elongate droplets (lower aspect ratio), while cluster 1 contains droplets that are more spherical. In contrast, clusters 2 and 3 contain fewer but larger structures, with cluster 3 showing the greatest mean volume. Slip velocities do not vary drastically among clusters, suggesting that size and shape features contribute more toward the similarity or difference of droplets' respective DIWECT values.

Next, we increase the number of clusters to analyze. This will provide further analysis of how the DIWECT differentiates physical features. In the $K = 4$ clusters, it was clear that the size and shape of droplets were ultimately the key features in differentiating them (i.e. see Figs. 11a and 11b). While these are key factors in atomization, velocity contributes significantly to breakup mechanisms. The effects of velocity should become more apparent, as there are enough clusters to capture droplets with similar shapes, but differing local velocity fields. This will provide further confidence that the WECT can successfully differentiate structures by not only geometry, but also by their surrounding flow conditions.

The results for $K = 16$ are shown in Fig. 12. The large cluster 1 in the original $K = 4$ collection is now spread out among clusters 1, 9, and 14, which display small-volume droplets with relatively high aspect ratios (see Fig. 13 for

visual confirmation). Among these, cluster 14 exhibits a lower mean aspect ratio compared to clusters 1 and 9 (≈ 0.5 vs. ≈ 0.6). While clusters 1 and 9 appear geometrically similar, they were clustered separately, likely suggesting an influence from velocity. Notably, cluster 9 displays a stronger x-direction slip velocity than cluster 1 ($\approx 35\text{m/s}$ vs. $\approx 25\text{m/s}$), suggesting that droplets in this grouping were subjected to differing flow conditions, potentially experiencing higher shear. This differentiation implies that variations in velocity fields contribute to different breakup behaviors, which are now captured by the greater number of clusters.

To quantitatively evaluate whether the clusters derived from the DIWECT display distinct physical characteristics, we performed a Permutational Multivariate Analysis of Variance (PERMANOVA) Anderson (2001). It is important to note that the physical quantities used in this test (volume, aspect ratio, and slip velocities in the x-, y-, and z-directions) are computed directly from the simulation and are not inputs into the DIWECT or to the clustering algorithm. The clusters are defined exclusively by distances in DIWECT-space. Statistically significant separation of the clusters with respect to the physical quantities therefore provides an external validation that the DIWECT captures physically meaningful information. PERMANOVA is nonparametric and determines whether clusters differ significantly based on their multivariate centroids. In this case volume, aspect ratio, and slip velocities (x,y, and z directions) were used. Conceptually similar to a traditional Analysis of Variance (ANOVA), PERMANOVA does not assume normality and instead computes variance from distances between samples in a distance matrix, allowing for multivariate comparisons. The test outputs a pseudo-F statistic, which mirrors the F-statistic in ANOVA as the ratio of variation between versus within groups. The accompanying p-value measures the probability that the F-statistic result could be obtained due to random chance. Thus, a low p-value (<0.001) indicates the observed differences are unlikely to be due to random noise.

Tests were performed on a random subsample of 10,000 droplets with 5,000 permutations. The independent variables used in the PERMANOVA analysis were volume, aspect ratio, and the x-, y-, and z-direction slip velocities. For $K = 4$ and $K = 16$, the pseudo- F values were approximately 450 and 150, respectively, with $p < 0.001$ in both cases. These results provide confidence that the clusters generated by the DIWECT descriptions are statistically separate populations when analyzed independently through the given variables. To evaluate the relative effects of geometry and the local flow field on the DIWECT, the tests were repeated on two subsets of the variables: a shape subset (volume and aspect ratio) and a velocity subset (the three slip velocity components). These subset tests all returned $p < 0.001$. The shape subset produced significant separation at both cluster resolutions (pseudo- $F \approx 1200$ for $K = 4$ and ≈ 470 for $K = 16$), which shows geometry to be a dominant feature. The velocity subset results additionally demonstrates that clusters can be statistically separated by their local flow fields alone, albeit to a lesser extent than geometry. The velocity subset produced pseudo- F values of ≈ 10 and ≈ 50 for $K = 4$ and $K = 16$, respectively. This trend indicates that with few clusters, the primary distinguishing feature is geometry, and as the number of clusters increases, velocity has a larger contribution. This quantitatively supports the qualitative observation that clusters 1, 9, and 14 (Fig. 13) contain geometrically similar droplets, which can be distinguished by their slip velocities. The slip velocity is a related, but distinct quantity from the interface-normal velocity used in the DIWECT. Without the velocity information available, it is unlikely that clusters with similar geometries would be distinguishable. The separation we observe in the clusters, along with the statistical analysis of the PERMANOVA provides confidence that the weighting in the DIWECT provides vital information for physical characteristic of droplets in different flow fields. A direct comparison of the weighting's individual contribution, obtained by comparing the normal velocity weighting to other potential velocity field weights is identified as future work.

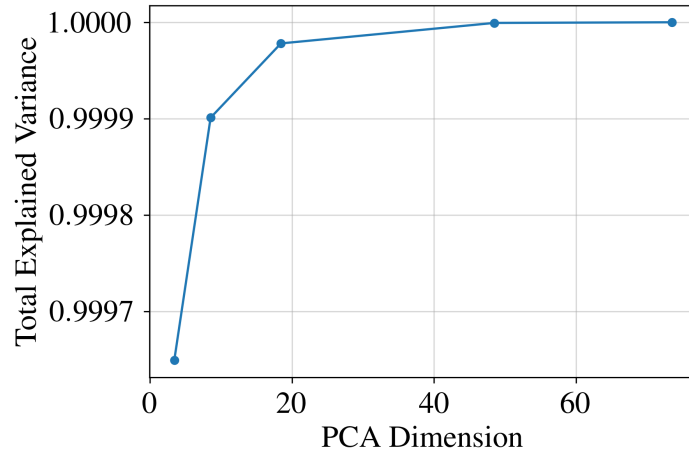
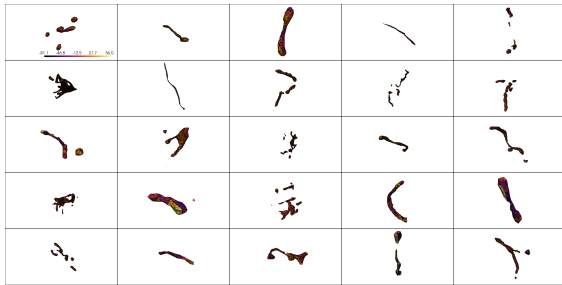
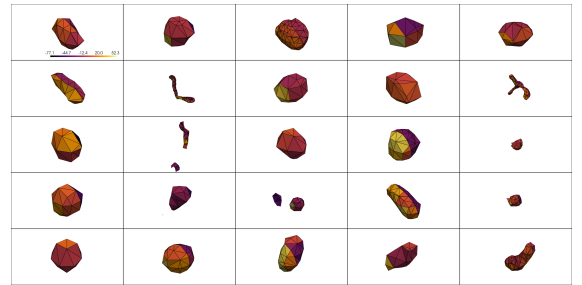


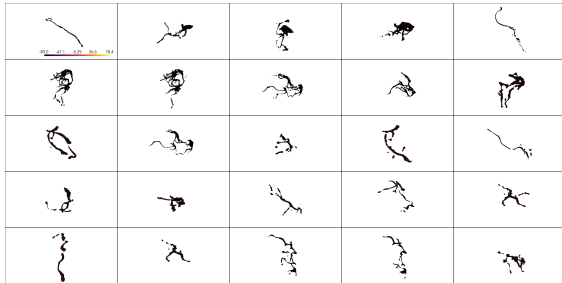
Figure 9: Total displayed variance of data with varying PCA dimensions.



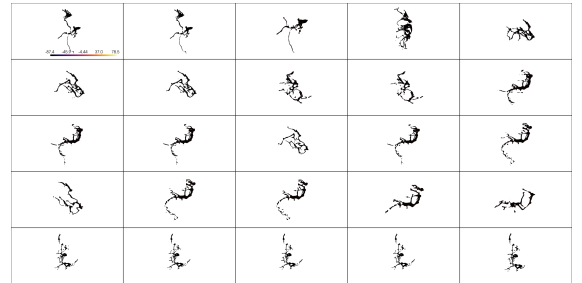
(a) K-means cluster 0.



(b) K-means cluster 1.

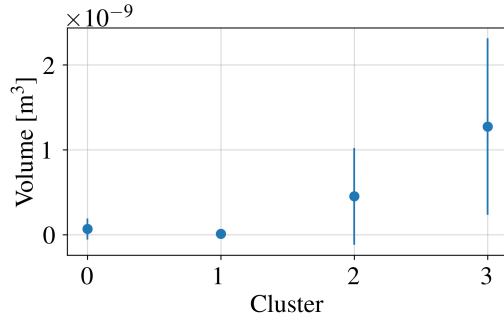


(c) K-means cluster 2.

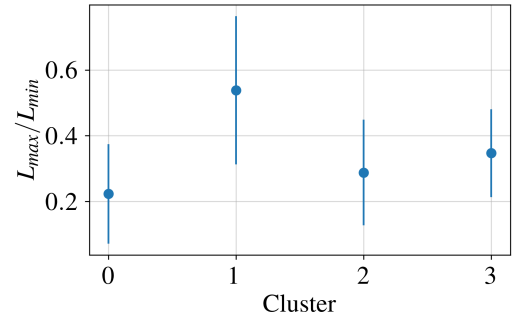


(d) K-means cluster 3.

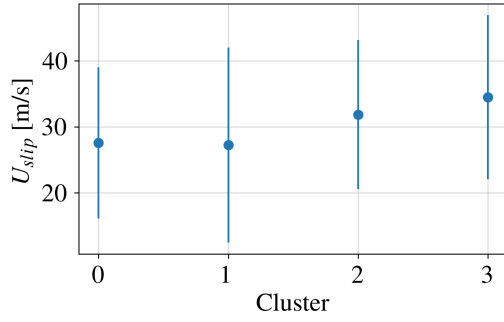
Figure 10: Visualization of the droplets in K-means clusters for $K = 4$.



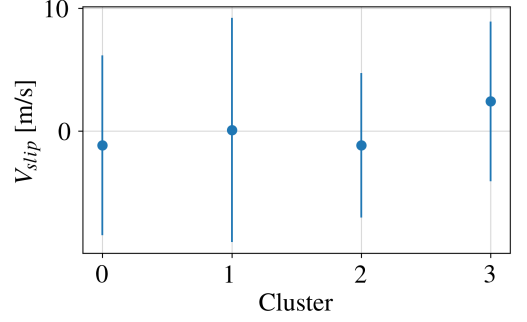
(a) Mean volume



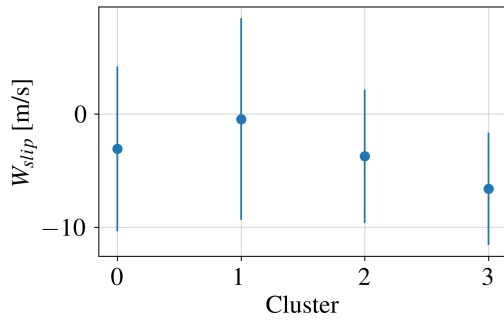
(b) Mean aspect ratio



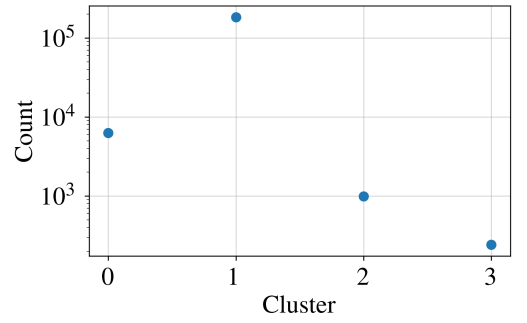
(c) Mean x-direction velocity



(d) Mean y-direction velocity



(e) Mean z-direction velocity



(f) Number of droplets

Figure 11: Statistics for clusters created by k-means clustering.

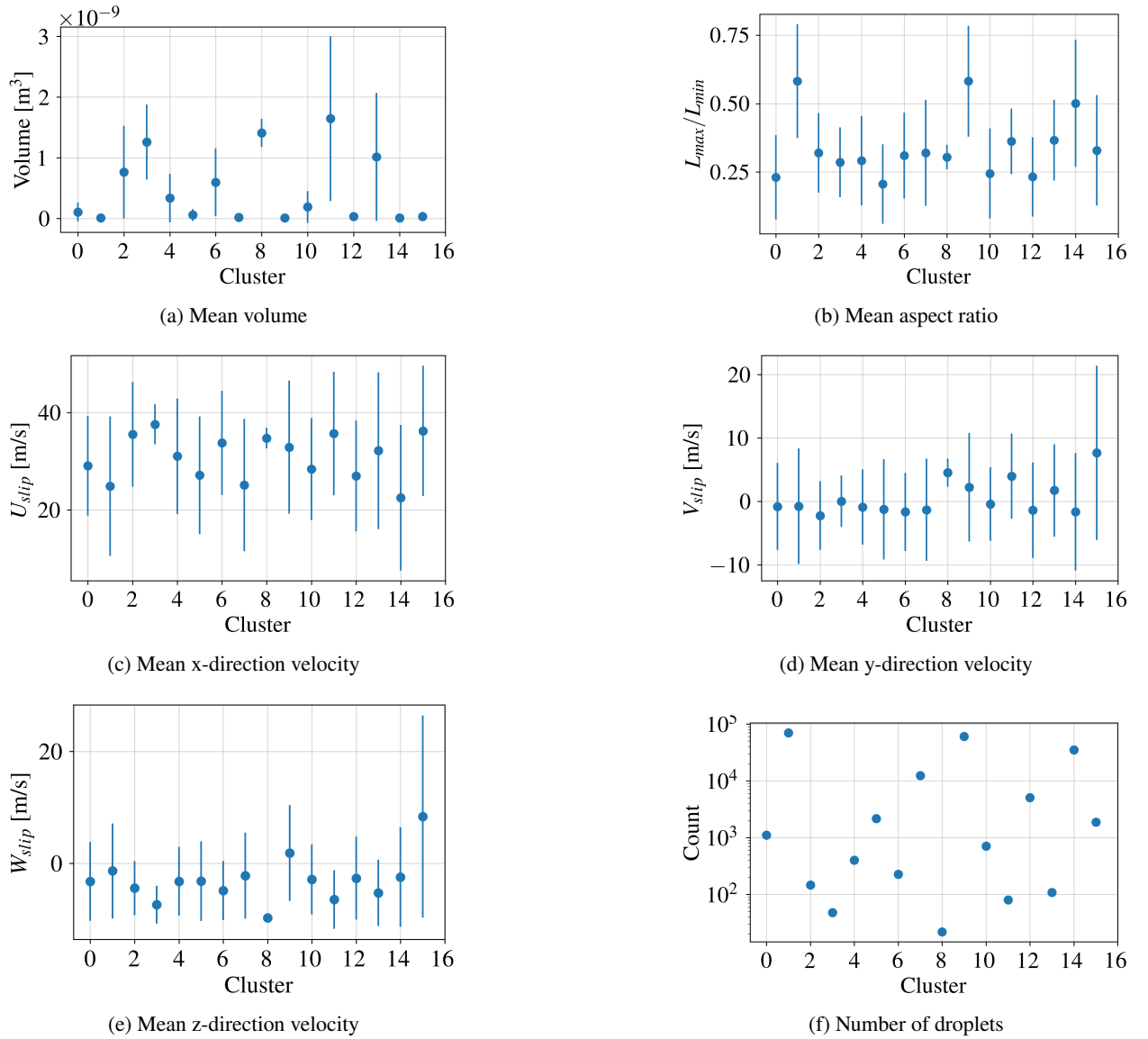


Figure 12: Statistics for clusters created by k-means clustering.

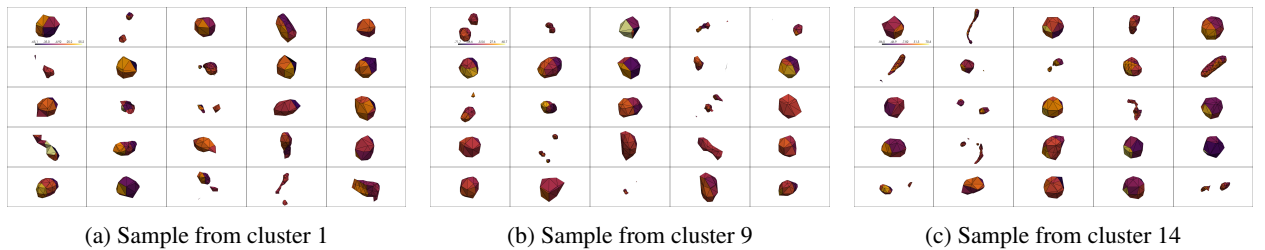


Figure 13: Renderings of samples of droplets from clusters 1, 9, and 14 in $K = 16$ clustering output.

5. Conclusions

This work has taken two significant steps toward a more comprehensive analysis of atomizing flows. First, we have shown that it is possible to extract liquid shapes and associated flow fields from high-fidelity simulations in a way that directly targets breakup and coalescence events. Previous approaches generally rely on full-field outputs Desjardins and Pitsch (2010); Shinjo and Umemura (2010); Zandian, Sirignano and Hussain (2017); Hasslberger et al. (2019a), but the method presented here isolates each relevant structure and captures localized information at the moment of breakup. This capability represents a shift from broad snapshots of the entire simulation domain to data that provides insights into model-relevant dynamics.

Second, we have demonstrated a means of transforming these complex, high-dimensional datasets into vectorized representations using the DIWECT. By approximating a mapping of geometry and local velocity fields to concise vectorized descriptors, we can readily apply machine learning or other data-driven tools to an otherwise intractable volume of simulation data. Our clustering examples illustrate that droplets sharing similar shape, size, and flow conditions appear in common groups, reaffirming that the DIWECT captures physically meaningful features.

While these contributions move atomization research closer to fully data-driven reduced-order modeling, there are limitations that must be addressed in future work. [First, an assessment of all possible weighting parameters would benefit the field. The present weighting uses only the interface-normal velocity as contribution of the flow field to the DIWECT. The contribution of this weighting relative to other potential scalar flow descriptors is an open question and area of potential future work.](#)

Furthermore, ASSET's requirement that distinct liquid structures be separated by a full grid cell can sometimes cause the output geometries to misrepresent connected components, either by grouping individual structures together or artificially separating continuous ones. This issue is evident in Figs. 10 and 13 and presents a challenge for the DIWECT, which is sensitive to discontinuities in interface representation. This artifact can be reduced with increasing simulation resolution, which is possible with continually improving numerical methods and computational efficiency. However, more robust treatment of liquid structures at the moment of breakup would improve the methodology. Implementations of numerical methods such as reconstruction with 2 planes (R2P) Han, Chiodi and Desjardins (2024) in output simulations would greatly improve these artifacts. [Ultimately, a database intended for predictive model development must be constructed from sufficiently resolved simulations. Because the extraction and vectorization methodology is independent of the input simulation, continuing improvements in resolution and numerical methods will lead to improved databases without modifying this approach.](#)

Despite the limitations, these methods show promise that physical data can be recovered from the DIWECT. As noted in §3.3, given an infinite number of sampling directions and heights, the DIWECT is a one-to-one mapping of any shape. Practically, the method must be discretized, so we chose 150 directions and 75 heights for the present application, then an estimate of the DIWECT was computed. This resulted in a data reduction from 133 GB to 10 MB. Applying the methodology presented in this work to a predictive tool will necessitate tuning these sampling parameters, however this remains outside the scope of the present work.

The key results in this work display the usefulness of the methods presented here. In §4.2 we show that the DIWECT can be used to find the closest match to a given droplet in a database. This provides a practical path toward a breakup model where simulation databases can be used to predict droplet dynamics. Additionally, §4.3 showed that these methods allow us to characterize all droplets in a given spray accurately and efficiently. These two improvements are novel, and useful in the field.

Building on this research and taking advantage of the vectorized representations of three-dimensional liquid shapes, further machine-learning applications can be applied. A regression study could be carried out with the DIWECT vectors as input features to predict continuous quantities associated with breakup, such as breakup time, child droplet size distributions, shape statistics, or local flow-field characteristics. Additionally, we highlighted an unsupervised clustering method, but future supervised classification methods could be employed. Training a model to recognize the DIWECT signatures of canonical breakup modes would allow for classification of all mechanisms responsible for atomization in any simulation and allow for a study of their distribution. Altogether, the transformation of raw simulation data into consistent vectors and statistics is a practical step toward enhanced data analysis and reduced-order modeling in computational atomization studies.

Ultimately, this work bridges the gap between massive simulation datasets and meaningful data for model development. The methods outlined here represent significant improvements in the field of atomization simulations,

by allowing greater utilization of massive, costly simulation runs. Future work will focus on model development to create a novel category of physics-based models without significant empirical assumptions.

Acknowledgments

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References

- Afanador, A.R., Zaleski, S., Tryggvason, G., Lu, J., 2021. Effect of topology changes on the breakup of a periodic liquid jet. *Computers & Fluids* 228, 105059.
- Amsden, A., 1997. KIVA3V. A Block-Structured KIVA Program for Engines with Vertical or Canted Valves. Technical Report. Los Alamos National Lab.(LANL), Los Alamos, NM (United States).
- Amsden, A., Ramshaw, J., O'Rourke, P., Dukowicz, J., 1985. KIVA: A computer program for two-and three-dimensional fluid flows with chemical reactions and fuel sprays. Technical Report. Los Alamos National Lab., NM (USA).
- Amsden, A.A., 1993. KIVA3. A KIVA program with block-structured mesh for complex geometries. Technical Report. Los Alamos National Lab.(LANL), Los Alamos, NM (United States).
- Amsden, A.A., O'Rourke, P.J., Butler, T.D., 1989. KIVA-II: A computer program for chemically reactive flows with sprays. Technical Report. Los Alamos National Lab.(LANL), Los Alamos, NM (United States).
- Anderson, M.J., 2001. A new method for non-parametric multivariate analysis of variance. *Austral Ecology* 26, 32–46. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1442-9993.2001.01070.pp.x>, doi:<https://doi.org/10.1111/j.1442-9993.2001.01070.pp.x>, arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1442-9993.2001.01070.pp.x>.
- Beale, J.C., Reitz, R.D., 1999. Modeling spray atomization with the kelvin-helmholtz/rayleigh-taylor hybrid model. *Atomization and Sprays* 9. doi:10.1615/atomizspr.v9.i6.40.
- Chen, X., Lu, J., Zaleski, S., Tryggvason, G., 2022. Characterizing interface topology in multiphase flows using skeletons. *Physics of Fluids* 34.
- Childs, H., Brugger, E., Whitlock, B., Meredith, J., Ahern, S., Pugmire, D., Biagas, K., Miller, M., Harrison, C., Weber, G.H., Krishnan, H., Fogal, T., Sanderson, A., Garth, C., Bethel, E.W., Camp, D., Rübel, O., Durant, M., Favre, J.M., Navrátil, P., 2012. Visit: An end-user tool for visualizing and analyzing very large data, in: *High Performance Visualization—Enabling Extreme-Scale Scientific Insight*, pp. 357–372. doi:10.1201/b12985.
- Choi, H., Moin, P., 1994. Effects of the computational time step on numerical solutions of turbulent flow. *Journal of Computational Physics* 113. doi:10.1006/jcph.1994.1112.
- Christensen, B., Owkes, M., 2023. Efficient extraction of atomization processes from high-fidelity simulations. *Computers and Fluids* 254. doi:10.1016/j.compfluid.2023.105808.
- Cisewski-Kehe, J., Fasy, B.T., Giryani, D., Quist, E., 2023. The weighted euler characteristic transform for image shape classification. arXiv preprint arXiv:2307.13940.
- Crawford, L., Monod, A., Chen, A.X., Mukherjee, S., Rabadán, R., 2020. Predicting clinical outcomes in glioblastoma: an application of topological and functional data analysis. *Journal of the American Statistical Association* 115, 1139–1150.
- Dendy, J.E., 1982. Black box multigrid. *Journal of Computational Physics* 48. doi:10.1016/0021-9991(82)90057-2.
- Desjardins, O., Blanquart, G., Balarac, G., Pitsch, H., 2008a. High order conservative finite difference scheme for variable density low mach number turbulent flows. *Journal of Computational Physics* 227, 7125–7159. doi:10.1016/j.jcp.2008.03.027.
- Desjardins, O., McCaslin, J.O., Owkes, M., Brady, P., 2013. Direct numerical and large-eddy simulation of primary atomization in complex geometries. *Atomization and Sprays* 23. doi:10.1615/AtomizSpr.2013007679.
- Desjardins, O., Moureau, V., Pitsch, H., 2008b. An accurate conservative level set/ghost fluid method for simulating turbulent atomization. *Journal of Computational Physics* 227, 8395–8416. doi:10.1016/j.jcp.2008.05.027.
- Desjardins, O., Pitsch, H., 2010. Detailed numerical investigation of turbulent atomization of liquid jets. *Atomization and Sprays* 20, 311–336.
- Dorschner, B., Biasiori-Poulanges, L., Schmidmayer, K., El-Rabii, H., Colonius, T., 2020. On the formation and recurrent shedding of ligaments in droplet aerobreakup. *Journal of Fluid Mechanics* 904, A20. URL: https://www.cambridge.org/core/product/identifier/S0022112020006990/type/journal_article, doi:10.1017/jfm.2020.699.
- Evrard, F., Denner, F., van Wachem, B., 2019. A multi-scale approach to simulate atomisation processes. *International Journal of Multiphase Flow* 119, 194–216.
- Fedkiw, R.P., Aslam, T., Merriman, B., Osher, S., 1999. A non-oscillatory eulerian approach to interfaces in multimaterial flows (the ghost fluid method). *Journal of Computational Physics* 152. doi:10.1006/jcph.1999.6236.
- Gameiro, M., Mischaikow, K., Wanner, T., 2005. Evolution of pattern complexity in the cahn–hilliard theory of phase separation. *Acta Materialia* 53, 693–704.
- Garcia, S., Ling, Y., 2026. Characterizing drop morphology evolution in aerodynamic breakup with topological skeleton. *Computers & Fluids* 304, 106894. URL: <https://www.sciencedirect.com/science/article/pii/S0045793025003548>, doi:<https://doi.org/10.1016/j.compfluid.2025.106894>.
- Gorokhovski, M., Herrmann, M., 2008. Modeling primary atomization. *Annual Review of Fluid Mechanics* 40, 343–366. doi:10.1146/annurev.fluid.40.111406.102200.
- Han, A., Chiodi, R., Desjardins, O., 2024. Capturing thin structures in vof simulations with two-plane reconstruction. *Journal of Computational Physics* 519, 113453.
- Hasslberger, J., Ketterl, S., Klein, M., Chakraborty, N., 2019a. Flow topologies in primary atomization of liquid jets: A direct numerical simulation analysis. *Journal of Fluid Mechanics* 859, 819–838. doi:10.1017/jfm.2018.845.
- Hasslberger, J., Marten, S., Klein, M., 2019b. A theoretical investigation of flow topologies in bubble- and droplet-affected flows. *Fluids* 4. doi:10.3390/fluids4030117.
- He, T., Hong, L., Chen, D., Liang, Z., 2001. Reliable path for virtual endoscopy: Ensuring complete examination of human organs. *IEEE transactions on visualization and computer graphics* 7, 333–342.
- Heindel, T.J., 2018. X-ray imaging techniques to quantify spray characteristics in the near field. *Atomization and Sprays* 28. doi:10.1615/ATOMIZSPR.2019028797.
- Herrmann, M., 2010. A parallel eulerian interface tracking/lagrangian point particle multi-scale coupling procedure. *Journal of Computational Physics* 229, 745–759. doi:10.1016/j.jcp.2009.10.009.

- Hilaga, M., Shinagawa, Y., Kohmura, T., Kunii, T.L., 2001. Topology matching for fully automatic similarity estimation of 3d shapes, in: Proceedings of the 28th annual conference on Computer graphics and interactive techniques, pp. 203–212.
- Hotz, I., Masood, T.B., Sadlo, F., Tierny, J., 2021. Topological methods in data analysis and visualization VI. Springer.
- Jarrahbashi, D., Sirignano, W.A., 2014. Invited article: Vorticity dynamics for transient high-pressure liquid injection. *Physics of Fluids* 26. doi:10.1063/1.4895781.
- Jiang, Q., Kurttek, S., Needham, T., 2020. The weighted euler curve transform for shape and image analysis, in: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), pp. 3685–3694. doi:10.1109/CVPRW50498.2020.00430.
- Kramár, M., Levanger, R., Tithof, J., Suri, B., Xu, M., Paul, M., Schatz, M.F., Mischaikow, K., 2016. Analysis of kolmogorov flow and rayleigh–bénard convection using persistent homology. *Physica D: Nonlinear Phenomena* 334, 82–98.
- Krishan, K., Kurtuldu, H., Schatz, M.F., Gameiro, M., Mischaikow, K., Madruga, S., 2007. Homology and symmetry breaking in rayleigh–bénard convection: Experiments and simulations. *Physics of Fluids* 19.
- Krolick, W., Owkes, M., 2018. Primary atomization instability extraction using dynamic mode decomposition. *Atomization and Sprays* 28, 1061–1079.
- Kuo, C.W., Trujillo, M.F., 2022. Simulation of liquid jet atomization and droplet breakup via a volume-of-fluid lagrangian–eulerian strategy. *Physics of Fluids* 34.
- Levanger, R., 2017. A Comparison Framework For Interleaved Persistence Modules and Applications of Persistent Homology to Problems in Fluid Dynamics. Ph.D. thesis.
- Lewiner, T., Lopes, H., Vieira, A.W., Tavares, G., 2003. Efficient implementation of marching cubes’ cases with topological guarantees. *Journal of graphics tools* 8, 1–15.
- Li, Z., Zhao, W., Wu, Z., 2021. Understanding transient internal flow processes in high-pressure nozzles using synchrotron radiation x-ray phase contrast imaging technology. *Atomization and Sprays* 31. doi:10.1615/ATOMIZSPR.2021035808.
- Linne, M.A., Paciaroni, M., Gord, J.R., Meyer, T.R., 2005. Ballistic imaging of the liquid core for a steady jet in crossflow. *Applied Optics* .
- Lloyd, S., 1982. Least squares quantization in pcm. *IEEE transactions on information theory* 28, 129–137.
- Lorensen, W., Cline, H., 1987. Marching cubes: A high resolution 3d surface construction algorithm. *Computer graphics* 21, 7–12.
- Marmottant, P.H., Villersmaux, E., 2004. On spray formation. *Journal of Fluid Mechanics* 498, 73–111. doi:10.1017/S0022112003006529.
- Meng, J.C., Colonius, T., 2018. Numerical simulation of the aerobreakup of a water droplet. *Journal of Fluid Mechanics* 835, 1108–1135. URL: https://www.cambridge.org/core/product/identifier/S0022112017008047/type/journal_article, doi:10.1017/jfm.2017.804.
- Nauleau, F., Bridel-Bertomeu, T., Vivodtzev, F., Beaugendre, H., Tierny, J., 2021. Topological data analysis for numerical method comparisons of 2d turbulent flows, in: Journée Visu 2021.
- Núñez, J., González, A., Ramos, E., 2022. Topological data analysis of lagrangian orbits in natural convection flows confined in a cylinder. *Physical Review Fluids* 7, 123501.
- O’Rourke, P., Amsden, A.A., 1987. The tab method for numerical calculation of spray droplet breakup. *SAE Technical Papers* doi:<https://doi.org/10.4271/872089>.
- Owkes, M., Cauble, E., Senecal, J., Currie, R.A., 2018. Importance of curvature evaluation scale for predictive simulations of dynamic gas–liquid interfaces. *Journal of Computational Physics* 365, 37–55. doi:10.1016/j.jcp.2018.03.018.
- Owkes, M., Desjardins, O., 2014. A computational framework for conservative, three-dimensional, unsplit, geometric transport with application to the volume-of-fluid (vof) method. *Journal of Computational Physics* 270, 587–612. doi:10.1016/j.jcp.2014.04.022.
- Owkes, M., Desjardins, O., 2017. A mass and momentum conserving unsplit semi-lagrangian framework for simulating multiphase flows. *Journal of Computational Physics* 332, 21–46. doi:10.1016/j.jcp.2016.11.046.
- Patterson, M.A., Reitz, R.D., 1998. Modeling the effects of fuel spray characteristics on diesel engine combustion and emission. *SAE Technical Papers* doi:10.4271/980131.
- Pilch, M., Erdman, C.A., 1987. Use of breakup time data and velocity history data to predict the maximum size of stable fragments for acceleration-induced breakup of a liquid drop. *Int. J. Multiphase Flow* 13, 741–757.
- Pilliod, J.E., Puckett, E.G., 2004. Second-order accurate volume-of-fluid algorithms for tracking material interfaces. *Journal of Computational Physics* 199, 465–502. doi:10.1016/j.jcp.2003.12.023.
- Pizer, S.M., Gerig, G., Joshi, S., Aylward, S.R., 2003. Multiscale medial shape-based analysis of image objects. *Proceedings of the IEEE* 91, 1670–1679.
- Reitz, R.D., Diwakar, R., 1987. Structure of high-pressure fuel sprays. *SAE Technical Papers* doi:10.4271/870598.
- Roy, A., Haque, R., Mitra, A., Tarafdar, S., Dutta, T., 2023. Characterizing fluid dynamical systems using euler characteristic surface and euler metric. *Physics of Fluids* 35.
- Rubel, C., Owkes, M., 2019. Extraction of droplet genealogies from high-fidelity atomization simulations. *Atomization and Sprays* 29, 709–739.
- Sedarsky, D., Berrocal, E., Linne, M., 2010. Numerical analysis of ballistic imaging for revealing liquid breakup in dense sprays. *Atomization and Sprays* 20. doi:10.1615/AtomizSpr.v20.i5.30.
- Sheehy, P., Owkes, M., 2017. Numerical study of electric reynolds number on electrohydrodynamic (ehd) assisted atomization. *Atomization and Sprays* 27. doi:10.1615/atomizspr.2017016576.
- Shinjo, J., Umemura, A., 2010. Simulation of liquid jet primary breakup: Dynamics of ligament and droplet formation. *International Journal of Multiphase Flow* 36, 513–532. doi:10.1016/j.ijmultiphaseflow.2010.03.008.
- Slangen, P.R., Lauret, P., Heymes, F., Aprin, L., Lecysyn, N., 2016. High-speed imaging optical techniques for shockwave and droplets atomization analysis. *Optical Engineering* 55, 121706. doi:10.1117/1.oe.55.12.121706.
- Tomar, G., Fuster, D., Zaleski, S., Popinet, S., 2010. Multiscale simulations of primary atomization. *Computers & Fluids* 39, 1864–1874.
- Torres, D.J., Trujillo, M.F., 2006. Kiva-4: An unstructured ale code for compressible gas flow with sprays. *Journal of Computational Physics* 219, 943–975.

- Turner, K., Mukherjee, S., Boyer, D.M., 2014. Persistent homology transform for modeling shapes and surfaces. *Information and Inference: A Journal of the IMA* 3, 310–344.
- U.S. Department of Energy, 2006. Modeling of combustion processes improves engine efficiency. URL: <https://www.osti.gov/servlets/purl/845713-htwC2h/>.
- Wang, Y.J., Im, K.S., Fezzaa, K., Lee, W.K., Wang, J., Micheli, P., Laub, C., 2006. Quantitative x-ray phase-contrast imaging of air-assisted water sprays with high weber numbers. *Applied Physics Letters* 89. doi:10.1063/1.2358322.
- Youngs, D.L., 1982. *Numerical Methods in Fluid Dynamics*. Academic Press. chapter Time-Dependent Multi-material Flow with Large Fluid Distortion. pp. 273–285.
- Yu, H., Goldsworthy, L., Ghiji, M., Brandner, P.A., Garaniya, V., 2017. A parallel volume of fluid-lagrangian parcel tracking coupling procedure for diesel spray modelling. *Computers & Fluids* 150, 46–65.
- Zandian, A., Sirignano, W.A., Hussain, F., 2017. Planar liquid jet: Early deformation and atomization cascades. *Physics of Fluids* 29. doi:10.1063/1.4986790.
- Zandian, A., Sirignano, W.A., Hussain, F., 2018. Understanding liquid-jet atomization cascades via vortex dynamics. *Journal of Fluid Mechanics* 843, 293–354. doi:10.1017/jfm.2018.113.
- Zuzio, D., Estivalèzes, J.L., DiPierro, B., 2018. An improved multiscale eulerian–lagrangian method for simulation of atomization process. *Computers & Fluids* 176, 285–301.